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Expectations

X : RV $\quad x \in \mathcal{X} \subseteq \mathbb{R}$

Discrete	Continuous
PMF p_X	PDF f_X
$E[X] = \sum_{x \in \mathcal{X}} x\, p_X(x)$	$E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx$
$\in \mathbb{R}$	

Bounded support : $\mathcal{X}$: bounded

$E[X]$: exists

Ex: $X \sim \text{Bernoulli}(p)$

$X \sim \{0, 1\}$ with 0 w.p. $1-p$, 1 w.p. p

$E[X] = p$

$X \sim \text{Uniform}[0,1)$

$E[X] = \int_0^1 x \cdot 1 \cdot dx = \frac{1}{2}$

$X \sim \text{Uniform}([-2,-1] \cup [1,2])$

$E[X] = 0$

Unbounded support

	Discrete		Continuous
$E(X)$	$\sum_{x \in \mathcal{X}, x \geq 0} x\, p_X(x)$	$\sum_{x \in \mathcal{X}, x < 0} x\, p_X(x)$	
exists	finite	finite	
∞	∞	finite	
$-\infty$	finite	$-\infty$	
does not exist	∞	$-\infty$	

Ex: (1) $X \sim \text{Geom}(p)$ $\quad X \in \{1, 2, \ldots, k, \ldots\}$ $\quad (1-p)^{k-1} p$

$$E[X] = \sum_{k=1}^{\infty} k \cdot (1-p)^{k-1} \cdot p = 1/p$$

(2) $X \sim \text{Poisson}(\lambda)$ $\quad p_X(k) = e^{-\lambda}\frac{\lambda^k}{k!}$ $\quad k = 0, 1, 2, \ldots$

$$E[X] = \sum_{k=0}^{\infty} k \cdot e^{-\lambda}\frac{\lambda^k}{k!} = \lambda$$

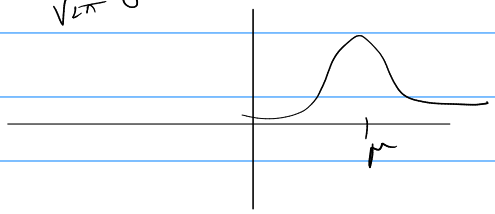
(3) $p_X(k) = \dfrac{1}{2|k|(|k|+1)}$, $k = \pm 1, \pm 2, \ldots$

$E[X]$: does not exist $\quad \sum_{k=1}^{\infty} \frac{1}{2(k+1)} = \frac{1}{2}\underbrace{\left(\frac{1}{2}+\frac{1}{3}+\cdots\right)}_{\infty}$

(4) $X \sim$ pdf $\lambda e^{-\lambda x}$, $x > 0$

$$E[X] = \int_0^{\infty} x \cdot \lambda e^{-\lambda x}\, dx = 1/\lambda$$

(5) $X \sim$ pdf $\dfrac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$, all x



$$E[X] = \mu$$

(6) $X \sim$ pdf $\dfrac{1}{\pi(1+x^2)}$ all x

$E[X]$: does not exist

limits are crazy

$1 + \frac{1}{2} + \frac{1}{3} + \cdots \to \infty$

✓ NOT ABSOLUTE

$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots \to \log 2$

$1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12}$

$+ \frac{1}{7} - \frac{1}{14} - \frac{1}{16} \cdots$

$= \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \cdots$

$= \frac{1}{2}\log 2$

Expectation of functions of RVs

X: RV $\quad g(X) \quad Y = g(X)$

Distn of Y

$E[g(X)] \qquad E[Y] = \sum_{y \in \mathcal{Y}} y\, p_Y(y)$

LOTUS (Law of the Unconscious statistician)

Discrete: $E[g(X)] = \sum_{x \in \mathcal{X}} g(x)\, p_X(x)$ "existence assumed"

Continuous: $E[g(X)] = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx$

Ex: $Y = X^2$ (2nd moment)

$E[X]$: 1st moment

$E[X^2]$: 2nd "

① $X \sim \text{Bernoulli}(p)$

$E[X] = p$

$E[X^n] = p$

② $X \sim \text{Uniform}[0,1]$

$E[X^2] = \frac{1}{3}$ $\qquad E[\cos 2\pi X] = 0$

$E[X^n] = \frac{1}{n+1}$

③ $X \sim \text{Geometric}(p)$

$E[X^2] = \sum_{k=1}^{\infty} k^2 (1-p)^{k-1} p = \frac{2}{p^2}$

④ $X \sim \text{Poisson}(\lambda)$

$E[X^2] = \sum_{k=0}^{\infty} k^2 \cdot e^{-\lambda} \frac{\lambda^k}{k!} = \lambda + \lambda^2$

⑤ $X \sim$ pdf $\lambda e^{-\lambda x}$, $x > 0$

$E[X^2] = \int_0^{\infty} x^2 \cdot \lambda e^{-\lambda x}\, dx \overset{?}{=} \frac{2}{\lambda^2}$

⑥ $X \sim$ pdf $\frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$, all x

$E[X^2] = \mu^2 + \sigma^2$

Multiple RVs $\quad X_1, X_2, \ldots, X_n$: joint distribution

$E[g(X_1, X_2, \ldots, X_n)]$